

Sindhu Vadapalli

REDACTED | isindhu925@gmail.com | [linkedin.com/in/sindhuvadapalli](https://www.linkedin.com/in/sindhuvadapalli) | github.com/prpl-25 | sindhu-vadapalli.vercel.app/

EDUCATION

Arizona State University <i>Master of Science in Computer Science: with thesis</i>	Expected Dec 2026 Tempe, AZ
Arizona State University <i>Bachelor of Science in Computer Science: Minor in Statistics</i>	Aug 2021 – May 2025 Tempe, AZ

EXPERIENCE

Software Engineering Intern <i>Capital One</i> Card Tech	June 2026 – Present Richmond, VA
Research Assistant <i>ASU Biodesign Institute</i> - Master's thesis research on reinforcement learning for personalized cancer therapy dosing policies using CancerSim, an agent based simulator. - Comparing the learned policy to adaptive therapy routines. Using cleanRL's ppo implementation on a custom gymnasium environment.	Oct 2025 – Present Tempe, AZ
Software Engineering Intern <i>Hidden Gemz</i> - Designed a collaborative filtering recommendation engine using Pandas, NumPy, and Scikit-learn to deliver real-time personalized suggestions. - Integrated the Google Maps API to calculate distance and travel time based on user location and transportation mode. Streamlined API and PostgreSQL database interactions to reduce latency and improve scalability. - Used GitLab for version control and Agile development practices, collaborating via Taiga storyboards to track user stories and tasks across a 4-person capstone group.	Aug 2024 – May 2025 Remote
Research Assistant <i>Self-Organizing Particle Systems Lab, ASU</i> - Developed and evaluated reinforcement learning rollout algorithm heuristics in Python to find near-optimal solutions to the NP-hard Symmetric and Asymmetric Traveling Salesman Problem (TSP and ATSP). - Implemented and compared an auction-based heuristic (AWPC) to the greedy nearest-neighbor heuristic across 70+ TSPLIB instances, achieving a 2.5% average cost improvement in 85% of cases of TSP cases. - Extended AWPC with epsilon scaling. Improving average cost improvement to 3.5% on TSP and 4.8% on ATSP over the nearest neighbor baseline - Visualized performance trends and comparative outcomes using Matplotlib and Pandas to support analysis.	Sep 2024 – Dec 2025 Tempe, AZ

PROJECTS

FreshB4 <i>React native, Gemini API, firestore</i> - Built a full-stack React Native mobile application that allows users to upload or capture picture, and view freshness analyses. - Leveraged Gemini API to integrate an LLM in our app to perform the freshness analysis, give the user recommendations, and detailed recipes. Won "Best use of Gemini-API" award at Sunhacks 2025.	Sep 2025
Cookin: your AI chef buddy! <i>React native, RN Executorch, LLaMA 3 model</i> - Developed an AI-powered cooking app using React Native and Executorch to simplify meal planning. - Deployed an on-device LLaMA 3 model to generate curated recipes based on user input. Integrated speech-to-text (STT) using react-voice, enhancing accessibility for diverse user needs. - Prompt engineered to fine-tune recipe quality, balancing accuracy and creativity. Published to the App Store with optimized performance and responsive UI, resulting in 15+ organic downloads.	July 2025

TECHNICAL SKILLS AND AWARDS

Languages: Java, Python, C/C++, SQL, C#, JavaScript, HTML/CSS, Swift
Frameworks: React, Node.js, Flask, JUnit, PyTorch, React Native, .NET
Developer Tools: Git, Jupyter Notebook, VS Code, Visual Studio, Eclipse
Awards: Dean's List, FURI award, National Science Foundation REU award, ASU NAMU - Provost's Award